3.3 Derivatives of Inverse Functions

You missed this lesson. Copy this into your notes. That is all you need to do.

Question of the day

If you mirror a curve across y = x, what happens to its slope?

Targets

- I can say how the points and slopes of f and its inverse are linked.
- I can find the slope of the inverse at a point using the derivative of f.
- I can write the tangent line to an inverse at a point.

Concepts

- **Inverse points** ((a, b) on f → (b, a) on f^{-1}): An inverse swaps inputs and outputs. If f(a) = b, then $f^{-1}(b) = a$.
- **The mirror** (rise ? run): The graph of f^{-1} is the graph of f mirrored across y = x. Mirroring swaps rise and run.
- **Reciprocal slopes** ($(f^{-1})?(b) = 1 / f?(a)$): Slopes at mirrored points are reciprocals. Flat becomes vertical.
- **Where to evaluate** (f(a) = b → f?(a)): Find a first: solve f(a) = b. Put a into f?, not b.
- **Tangent to the inverse** ($y - a = (1 / f?(a))(x - b)$): The point is (b, a). The slope is 1 / f?(a).

What the class predicted, and what is true

- The line $y = 2x$ has slope 2. Mirror it across $y = x$. The new line has slope... **1/2.** The mirror swaps x and y: $x = 2y$, so $y = \frac{1}{2}x$. Rise and run trade places.
- $f(x) = x^3$ has a flat tangent at the origin. After mirroring across $y = x$, that tangent becomes... **vertical.** Rise and run swap. A run with no rise (flat) becomes a rise with no run (vertical).

Huddle notes (answers filled in)

Definitions

- **Inverse:** An inverse reverses **inputs** and outputs: if f(a) = b then $f^{-1}(b) =$ **a**.
- **Mirror:** The graph of f^{-1} is the graph of f reflected across **y = x**.
- **Slopes:** A reflection swaps rise and run, so the two slopes are **reciprocals**.

Worked example

$f(x) = x^3 + x$. Find $(f^{-1})?(2)$.

- Solve f(a) = 2. By inspection a = **1**
- The derivative is $3x^2 + 1$, so at x = 1 it is **4**
- Flip it: $(f^{-1})?(2) =$ **1/4**

Label the mirror

1. The curve through (1, 2) is the graph of **f**.
2. Its mirror through (2, 1) is the graph of $\mathbf{f^{-1}}$.
3. f has slope 4 at (1, 2), so f^{-1} has slope **1/4** at (2, 1).

One worked sample of each skill

Flip the slope

Level 1. $f(3) = 26$ and $f'(3) = 4$. Find $(f^{-1})'(26)$.

Answer: 0.25 The inverse slope is the reciprocal: $\dfrac{1}{4} = 0.25$.

Level 2. $f(4) = -7$ and $f'(4) = -10$. Find $(f^{-1})'(-7)$.

Answer: -0.1 The reciprocal keeps the sign: $\dfrac{1}{-10} = -0.1$.

Level 3. The table gives selected values of a one-to-one function f and its derivative. Find $(f^{-1})'(18)$.

Answer: 0.25 $f(2) = 18$, so use $f'(2) = 4$. The answer is $\dfrac{1}{4} = 0.25$.

Level 4. The table gives selected values of a one-to-one function f and its derivative. Find $(f^{-1})'(34)$.

Answer: 0.1 $f(3) = 34$, so use $f'(3) = 10$. The answer is $\dfrac{1}{10} = 0.1$.

Where does f? go

Level 1. $f(x) = x^3 + 2x$ and $f(1) = 3$. To find $(f^{-1})'(3)$ you evaluate f' at...
(A) x = 1 (B) x = 1/3 (C) x = 2 (D) x = 3
Answer: A $f(1) = 3$, so the input is 1. Evaluate $f'(1)$, then flip it.

Level 2. $f(x) = x^3 + x$ and $f(1) = 2$. To find $(f^{-1})'(2)$ you evaluate f' at...
(A) x = 2 **(B) x = 1** (C) x = 1/2 (D) x = 10
Answer: B $f(1) = 2$, so the input is 1. Evaluate $f'(1)$, then flip it.

Level 3. $f(x) = e^x + 3x$ and $f(0) = 1$. To find $(f^{-1})'(1)$ you evaluate f' at...
(A) x = 1 (B) x = 8 (C) x = 1/1 **(D) x = 0**
Answer: D $f(0) = 1$, so the input is 0. Evaluate $f'(0)$, then flip it.

Level 4. $f(x) = e^x + 3x$ and $f(0) = 1$. To find $(f^{-1})'(1)$ you evaluate f' at...
(A) x = 1/1 (B) x = 1 **(C) x = 0** (D) x = 8
Answer: C $f(0) = 1$, so the input is 0. Evaluate $f'(0)$, then flip it.

Slope of the inverse

Level 1. $f(x) = 5x + 4$. Find $(f^{-1})'(29)$.

Answer: 0.2 $f'(x) = 5$ everywhere, so $(f^{-1})' = \dfrac{1}{5} = 0.2$.

Level 2. $f(x) = x^3 + x$. Find $(f^{-1})'(2)$.

Answer: 0.25 $f(1) = 2$. $f'(1) = 4$. So $(f^{-1})'(2) = \dfrac{1}{4} = 0.25$.

Level 3. $f(x) = e^{3x} + x$. Find $(f^{-1})'(1)$.

Answer: 0.25 $f(0) = 1$. $f'(0) = 4$. So $(f^{-1})'(1) = \dfrac{1}{4} = 0.25$.

Level 4. $f(x) = e^{3x} + x$. Find $(f^{-1})'(1)$.

Answer: 0.25 $f(0) = 1$. $f'(0) = 4$. So $(f^{-1})'(1) = \dfrac{1}{4} = 0.25$.

Tangent to the inverse

Level 1. Write the tangent line to f^{-1} at the point $(3, 1)$.

Answer: $\frac{1}{5}(x-3)+1$ The point is $(3, 1)$ and the slope is $\dfrac{1}{5}$. So $y = \dfrac{1}{5}(x - (3)) + (1)$.

Level 2. Write the tangent line to f^{-1} at the point $(2, 1)$.

Answer: $\frac{1}{4}(x-2)+1$ The point is $(2, 1)$ and the slope is $\dfrac{1}{4}$. So $y = \dfrac{1}{4}(x - (2)) + (1)$.

Level 3. Write the tangent line to f^{-1} at the point $(1, 0)$.

Answer: $\frac{1}{4}(x-1)+0$ The point is $(1, 0)$ and the slope is $\dfrac{1}{4}$. So $y = \dfrac{1}{4}(x - (1)) + (0)$.

Level 4. Write the tangent line to f^{-1} at the point $(1, 0)$.

Answer: $\frac{1}{4}(x-1)+0$ The point is $(1, 0)$ and the slope is $\dfrac{1}{4}$. So $y = \dfrac{1}{4}(x - (1)) + (0)$.